

MID TERM EXAMINATION**B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)****B.Tech 3rd Semester — November, 2023****Paper Code: CIC-205 • Subject: Discrete Mathematics****Time:** 1½ Hrs.**Max. Marks:** 30**Note:** Q. No. 1 is compulsory and attempt any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	What is extended pigeonhole principle? Explain with suitable example.	2	CO1
(b)	Write down the converse, contrapositive, inverse and negation of the following sentence: “If it rains then the crop will grow.”	2	CO1
(c)	Represent the given statement using predicate and quantifier and negate it: “For all the real number x if $x > 5$ then $x^2 > 25$.”	2	CO1
(d)	What is the necessary condition for the relation to become Poset? Explain with example.	2	CO2
(e)	What is Function? Write the condition of the Function to be injective.	2	CO2
2. (a)	Consider these statements, of which the first three are premises and the fourth is a valid conclusion: • “All humming birds are richly colored.” • “No large birds live on honey.” • “Birds that do not live on honey are dull in color.” • “Humming birds are small.” Express the statements in the argument using quantifiers, assume that the domain consists of all birds.	5	CO1
(b)	Find the CNF and PCNF of $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$.	5	CO1
3. (a)	Let $A = \{1, 2, 3, 4, 6, 12\}$ and R is a Relation on the Set A such that aRb if a divides b. Find: (i) Relation R as a set of ordered pairs (ii) Digraph of R (iii) Find Adjacency matrix of R (iv) In-degree and out-degree of each node (v) Find its Hasse diagram	6	CO2
(b)	Solve the recurrence Relation using Master Method: $T(n) = T(n/2) + 2^n$	4	CO2
4. (a)	Consider the following five relations on the set $A = \{1, 2, 3, 4\}$: $R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$ $R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$ $R_3 = \{(1, 3), (2, 1)\}$ $R_4 = \emptyset$, the empty relation $R_5 = A \times A$, the universal relation Determine which of the relations are Reflexive, Symmetric, Antisymmetric, Asymmetric, Transitive and Equivalence.	6	CO1
(b)	What is the Generating Function for the sequence 1, 9, 25, 49, ... ?	4	CO2